



PRODUCT NAME

High-performance Dishwashing Liquid Formula

Description

Manual dishwashing detergent for kitchen use

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100-500 kg
Processing Time:	2-4 hours
Temperature:	Room temperature
Storage Conditions:	Cool, dry place
Equipment:	Standard mixer
Certification:	Meets industry standards

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Sodium Lauryl Ether Sulfate (SLES)	Sodium Laureth Sulfate	NaN%	Primary surfactant for cleaning and foaming
2	Cocamidopropyl Betaine	Cocamidopropyl Betaine	NaN%	Co-surfactant for foam boosting and mildness
3	Sodium Chloride	Sodium Chloride	NaN%	Viscosity adjuster
4	Ethoxylated Alcohol	C9-11 Pareth-8	NaN%	Non-ionic surfactant for grease cutting
5	Sodium Citrate	Sodium Citrate	NaN%	Builder for water softening
6	Citric Acid	Citric Acid	NaN%	pH adjuster
7	Preservative	Methylisothiazolinone and Methylchloroisothiazolinone	NaN%	Preservative to ensure product longevity
8	Colorant	Colorant	NaN%	Provides appealing appearance
9	Fragrance	Fragrance	NaN%	Imparts a pleasant scent
10	Water	Aqua	NaN%	Diluent and base for the formulation
Total			100%	

Manufacturing Process

Phase 1: Manufacturing Phase

Phase 2: Manufacturing Phase

Phase 3: Manufacturing Phase

Phase 4: Manufacturing Phase

Phase 5: Manufacturing Phase

Phase 6: Manufacturing Phase

Phase 7: Manufacturing Phase

Phase 8: Manufacturing Phase

Required Equipment

Standard mixer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Cool, dry place

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Meets industry standards

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100-500 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply as needed according to product instructions